

Multiple Parameters and Linear Regression

UBCO MDS — DATA 582





- ▶ Let's continue discussion of multiple parameters in Bayesian inference.
- ▶ So far we discussed the MH-approach for MCMC from a joint posterior, and introduced the Gibbs sampling for conditional posteriors
- ▶ Of course, these are approximation methods if, say, the joint/marginal posteriors are tricky to find/deal with
- ▶ Now we return to the normal distribution...

- ▶ If $X \sim N(\mu, \sigma^2)$ and both parameters are unknown, then we need to define a joint prior distribution $g(\mu, \sigma^2)$
- ▶ This already seems daunting! How can I propose a prior distribution that encapsulates my beliefs about both μ and σ^2 simultaneously?
- ▶ Remember, μ and σ^2 are merely random variables. Joint distributions are easy if we are
 - ▶ willing to assume that the random variables are independent

$$g(\mu, \sigma^2) = g_1(\mu)g_2(\sigma^2)$$

- ▶ able to define them conditionally

$$g(\mu, \sigma^2) = g_1(\mu)g_2(\sigma^2 \mid \mu) = g_1(\sigma^2)g_2(\mu \mid \sigma^2)$$

- ▶ The **joint posterior** distribution is

$$h(\mu, \sigma^2 \mid \mathbf{x}) \propto \phi(\mathbf{x} \mid \mu, \sigma^2)g(\mu, \sigma^2)$$

- ▶ This joint posterior could be used for joint inference directly (for example, $E[\theta]$ where θ is the vector (μ, σ^2) could be found/approximated).
- ▶ Often, we may want inference on the **marginal posterior** distribution of one (or more) of the parameters individually.
- ▶ As an example, the marginal posterior for μ would be found via

$$h(\mu \mid \mathbf{x}) = \int h(\mu, \sigma^2 \mid \mathbf{x})d\sigma^2$$



- ▶ Considering that $h(\mu, \sigma^2 \mid \mathbf{x}) \propto \phi(\mathbf{x} \mid \mu, \sigma^2)g(\mu, \sigma^2)$ has been our focus, a natural question is: do we need to be careful with g ?
- ▶ For example, what if we propose a function g which does not integrate to 1 (and therefore is not actually a probability distribution)?
- ▶ We call these **improper prior** distributions. It turns out these are okay for several inferential purposes so long as we end up with a posterior that integrates to something finite.
- ▶ Improper priors are generally used to specify uninformative priors. For example, suppose I want to specify a uniform prior for μ ?

- ▶ Let's assume the following
 - ▶ $X \sim N(\mu, \sigma^2)$
 - ▶ $g_1(\mu) \propto 1$ (improper)
 - ▶ $g_2(\sigma^2) \propto \frac{1}{\sigma^2}$ (improper, "Jeffrey's prior")
 - ▶ $g(\mu, \sigma^2) = g_1(\mu)g_2(\sigma^2) \propto (1)\frac{1}{\sigma^2}$

$$\begin{aligned}h(\mu, \sigma^2 \mid \mathbf{x}) &\propto f(\mathbf{x} \mid \mu, \sigma^2) \times p(\mu, \sigma^2) \\&\propto \frac{\exp \left\{ -\frac{1}{2\sigma^2} \left[n(\bar{x} - \mu)^2 + \sum_{i=1}^n (x - \bar{x})^2 \right] \right\}}{(\sigma^2)^{n/2}} \times \frac{1}{\sigma^2} \\&\propto \frac{1}{(\sigma^2)^{\frac{n}{2}+1}} \exp \left\{ -\frac{1}{2\sigma^2} \left[n(\bar{x} - \mu)^2 + (n-1)s^2 \right] \right\}\end{aligned}$$

where $s^2 = \frac{\sum_{i=1}^n (x - \bar{x})^2}{n-1}$ is the sample variance.

Normal Distribution: Marginal Posterior

- If we treat σ^2 as a nuisance parameter, then we are only interested in inference on μ

$$\begin{aligned} h(\mu | \mathbf{x}) &= \int h(\mu, \sigma^2 | \mathbf{x}) d\sigma^2 \\ &= \int h(\mu | \sigma^2, \mathbf{x}) h(\sigma^2 | \mathbf{x}) d\sigma^2 \end{aligned}$$

- It turns out that $h(\mu | \sigma^2, \mathbf{x})$ under the improper constant prior for μ is well-known:

$$\mu | \sigma^2, \mathbf{x} \sim N(\bar{x}, \sigma^2/n)$$

- We can find the marginal posterior for σ^2 :

$$\sigma^2 | \mathbf{x} \sim \text{Inv-Gamma}\left(\frac{n-1}{2}, \frac{(n-1)s^2}{2}\right)$$



- ▶ When we go to integrate out σ^2 , we find that $h(\mu \mid \mathbf{x})$ can be recognized as a generalized Student's t -distribution!

- ▶ In short

$$\mu \mid \mathbf{x} \sim t(\bar{x}, \frac{s^2}{n}, \nu = n - 1)$$

- ▶ Does this look familiar? It should...sort of...under frequentist framework

$$\bar{x} \mid \mu \sim t(\mu, \frac{s^2}{n}, \nu = n - 1)$$

Normal Distribution: Bayesian vs Frequentist



- ▶ In 'studentized' form, in other words
- ▶ Bayesian

$$\frac{\mu - \bar{x}}{\sqrt{s^2/n}} \sim t_{n-1}$$

- ▶ Frequentist

$$\frac{\bar{x} - \mu}{\sqrt{s^2/n}} \sim t_{n-1}$$

- ▶ We've effectively found an equivalency between the frequentist approach for μ under unknown σ^2 and the Bayesian approach for μ under unknown σ^2 (specifically using the priors we chose earlier).
- ▶ The difference is what we consider fixed (the observed data, or the parameter).



- ▶ Let's approach linear regression from a Bayesian framework.
- ▶ Recall that regression analysis seeks to explain or predict a *response variable* (outcome), Y based on one or more *predictor variables* X
- ▶ As its name suggests, the relationship between the response variable and the predictor should be (roughly) linear.
- ▶ We might use some transformation on the X variables to achieve this.

- ▶ The regression equation with one covariate is *simple* linear regression model given by

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad (1)$$

where

- ▶ β_0 is the intercept (the expected value of $Y \mid X = 0$)
- ▶ β_1 is the slope (the expected difference Y for one unit increase in X).
- ▶ ϵ are the independent errors following a $N(0, \sigma^2)$

β_0 and β_1 are referred to as *regression coefficients* and can be stored in a vector $\beta = (\beta_0, \beta_1)$.



- ▶ DATA 570 was focused on the frequentist approach for this model.
- ▶ There we derived the so-called OLS estimates for the regression coefficients which we denote $\hat{\beta}_j$.
- ▶ In the Bayesian treatment, we will arrive at a posterior distribution for the slope and intercept.
- ▶ These posterior distributions, as before, will contain all the uncertainty in our measurement, given our data.
- ▶ Point estimates, of course, can be obtained from the posterior as we have seen in other Bayesian examples.

- ▶ Bayesian simple linear regression is closely related to the normal models we've just considered
- ▶ However, rather than the observations having a common population mean μ , each observation has a mean dependent on the regression coefficients and the i th predictor.
- ▶ In notation we can write:

$$E(Y_i \mid x_i, \beta_0, \beta_1, \sigma^2) = \beta_0 + \beta_1 x_i$$

- ▶ For a simplified handling, let's consider centering the predictors so that our model becomes:

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i \quad (2)$$

where

- ▶ β_0 is the *centred* intercept (the expected value of $Y \mid X = \bar{X}$)
 - ▶ β_1 is the slope (the expected difference Y for one unit increase in X).
 - ▶ ϵ are the independent errors following a $N(0, \sigma^2)$
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- ▶ This transformation can improve the convergence MCMC algorithms that may be used to fit these models.

- Assuming centered predictors, we have

$$y_i \mid x_i, \beta_0, \beta_1, \sigma^2 \sim N(\beta_0 + \beta_1(x_i - \bar{x}), \sigma^2)$$

Hence our likelihood function could be written:

$$\begin{aligned} &= \prod_{i=1}^n \phi(y_i \mid x_i, \beta_0, \beta_1, \sigma^2) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y_i - (\beta_0 + \beta_1(x_i - \bar{x})))^2 \right\} \\ &= \frac{1}{(\sigma^2)^{n/2}} \exp \left\{ \frac{1}{2\sigma^2} \sum_i (y_i - \beta_0 - \beta_1(x_i - \bar{x}))^2 \right\} \end{aligned}$$



- ▶ Since we are primarily interested in slope, we would like to find a marginal posterior distribution for β_1 .
- ▶ To do this, we need to integrate out the nuisance parameters (β_0 and σ^2) from the joint posterior distribution.
- ▶ But first let's set our prior ...

- ▶ As we did for our normal model with μ and σ^2 unknown, we can treat the parameters as independent so that

$$g(\beta_0, \beta_1, \sigma^2) = g_1(\beta_0)g_2(\beta_1)g_3(\sigma^2)$$

- ▶ Using uniform (improper) priors for β_0 and β_1 , along with Jeffrey's (improper) prior for σ^2 we get:

$$g(\beta_0, \beta_1, \sigma^2) \propto 1 \times 1 \times \frac{1}{\sigma^2} = \sigma^{-2} \quad (3)$$

for $-\infty < \beta_0, \beta_1 < \infty$ and $0 < \sigma^2 < \infty$.

- To simplify some of the expressions to follow we adopt the following:

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} \\ \hat{\beta}_1 &= \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} \\ SSR &= \sum_i \left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x}) \right)^2\end{aligned}\tag{4}$$

where the sample variance can be expressed as $s^2 = SSR/(n-2)$.

- Thus, the joint posterior density is:

$$\begin{aligned}h(\beta_0, \beta_1, \sigma^2 \mid y) &\propto g(\beta_0, \beta_1, \sigma^2) \times \phi(y_i \mid x_i, \beta_0, \beta_1, \sigma^2) \\&= \frac{1}{\sigma^2} \frac{1}{(\sigma^2)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_i (y_i - \beta_0 - \beta_1(x_i - \bar{x}))^2 \right\} \\&= \frac{1}{(\sigma^2)^{\frac{n+1}{2}}} \exp \left\{ -\frac{\sum_i (y_i - \beta_0 - \beta_1(x_i - \bar{x}))^2}{2\sigma^2} \right\}\end{aligned}$$

- Note: this posterior will be ‘proper’ so long as $n > 2$ and predictor values are not all equal.

- ▶ Integrating out our nuisance parameters from the joint posterior on the previous page should give us the marginal posterior of β_1 .

$$h(\beta_1 | y) = \int_{-\infty}^{\infty} \int_0^{\infty} h(\beta_0, \beta_1, \sigma^2 | y) d\sigma^2 d\beta_0$$

- ▶ With a bit of work, we find that this has a functional form of a generalized t distribution:

$$\beta_1 | y \sim t \left(\hat{\beta}_1, \frac{s^2}{\sum (x_i - \bar{x})^2}, n - 2 \right)$$

- ▶ ...

- ▶ This prior specification yields Bayesian inference analogous to the frequentist results.
- ▶ For example, corresponding symmetric $c\%$ credible and confidence intervals will be computed as the same numbers for any observed sample.
- ▶ They are still interpreted differently (what's considered to be “randomly varying” and what's considered to be fixed), but it's perhaps reassuring that the two methods *can* somewhat agree on the end result.



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